

ROUTE2ZERO PILOT EXECUTION PLAN

Route2Zero Pilot Plan

A six-month, evidence-first pilot for equitable e-jeepney corridor prioritization in Metro Manila

PILOT INTENT

Move Route2Zero from a transparent historic-data MVP to a city-validated decision-support process that identifies which corridors are ready for deeper electrification planning, what evidence is still missing, and who owns each next action.

COVERAGE TARGET	EVIDENCE STANDARD	DECISION OUTPUT
3 spotlight LGUs	100% source-labeled	1 validated shortlist

Pilot objectives

Refresh the baseline	Confirm current route, service, operator, and depot evidence instead of treating the 2013-2020 feed as live service.
Co-design the ranking	Test criteria, weights, confidence language, and safeguards with city offices, operators, drivers, and rider/community representatives.
Validate feasibility	Screen shortlisted corridors for service continuity, depot and charging constraints, operator finance, and implementation dependencies.
Leave a maintainable tool	Hand over documented data, editable overrides, reproducible scripts, trained owners, and a prioritized next-phase roadmap.

Proposed pilot period: six months after award. Exact dates, partner commitments, and city scope will be confirmed during mobilization.

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Activities and execution timeline

Five gated stages connect data refresh, co-design, field validation, decision packaging, and handover.

Timeline	Activity / milestone	Outcome
Months 1-2	Mobilize, refresh, and agree Confirm pilot governance; inventory current route, fleet, ridership, settlement, depot, and grid evidence; agree validation protocol and safeguards.	Signed pilot charter; data inventory; GTFS refresh plan; ethics and risk register; workshop calendar.
Month 3	Co-design the evidence model Run city, operator, and rider sessions; test scoring definitions and weights; populate operator-readiness evidence; log contested assumptions.	Validated criteria and evidence rubric; operator override register; documented weight proposals; community issue log.
Month 4	Validate routes and charging feasibility Ground-check shortlisted corridors; verify stop order and service pattern; screen candidate depots, feeder constraints, tariffs, and charging operating windows.	Verified pilot-route geometry; depot/charging options matrix; updated confidence grades; routes removed or escalated with reasons.
Month 5	Calibrate and package decisions Re-run scoring and sensitivity scenarios; produce city shortlists, policy briefs, finance questions, and implementation dependencies.	Decision-ready shortlist; sensitivity report; policy briefs; financing and procurement question set; dashboard release candidate.
Month 6	Demonstrate, hand off, and decide scale Train city and partner users; conduct final review; compare outputs with stakeholder judgment; agree ownership and next-phase data sharing.	Pilot evaluation; trained owners; handover pack; public methodology; scale/no-scale decision and 12-month roadmap.

Decision gates

GATE 1 • END M2 Enough current evidence to begin co-design	GATE 2 • END M4 Shortlist survives route and depot validation	GATE 3 • END M6 Named owners and data-sharing path support scale
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Workstreams, deliverables, and acceptance

Each workstream has a named evidence boundary and a concrete handoff.

Workstream	Core activities	Deliverables	Acceptance test
A · Data and model	Refresh GTFS and reference layers; maintain provenance; run sensitivity analysis; protect nulls and proxy labels.	Versioned data inventory, reproducible pipeline, score cards, confidence ledger.	Every score traces to source, proxy, placeholder, or explicit missing state.
B · City and community validation	Facilitate LGU, operator, driver, and rider sessions; verify route realities and equity concerns; record disagreement.	Workshop records, validated issue map, equity evidence requests, agreed review criteria.	No corridor is called ‘equitable’ solely from WorldPop density.
C · Grid and depot feasibility	Screen candidate depots, charging windows, distribution constraints, tariffs, resiliency, and solar/storage pairing.	Depot options matrix, utility question list, interconnection risks, refinement backlog.	Regional grid share is never presented as local charger capacity.
D · Decision products and adoption	Publish city views, briefs, exportable rankings, training, governance, and scale decision pack.	Dashboard release, policy briefs, operator override workflow, handover and roadmap.	A city user can reproduce, challenge, export, and update the shortlist.

Success measures

Measure	Pilot target	Evidence
Data transparency	100% of scored fields retain source and confidence labels	Automated data audit and route-level export
Stakeholder validation	At least one structured validation cycle per spotlight LGU	Attendance, issues, decisions, and unresolved items log
Decision usability	Partner users can filter, challenge weights, export, and explain a shortlist	Observed task test and closing survey
Handover readiness	Named owners for data refresh, overrides, and public methodology	Signed handover and 12-month roadmap

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Team structure • Product and evidence

Role allocations define pilot accountability. Personal biographies and CV evidence can be appended if requested by the organizers.

Name	Pilot role	Relevant skills applied / accountable outputs
John Marwin Ebona	Project Lead and Policy Integration	Project coordination, decision framing, partner governance, risk escalation, and final pilot acceptance. Accountable for charter, decision gates, and scale recommendation.
Prince Marl	Lead Data Engineer	Data ingestion, validation, reproducibility, source controls, processed outputs, and operator-override workflow. Accountable for pipeline reliability and audit results.
Joaquin Sarmiento	Geospatial and Transport Analytics Lead	GTFS route logic, representative trips, geometry confidence, route-city tagging, field-validation protocol, and corridor map QA.
Isaac Marcus	Product and Streamlit Lead	Responsive dashboard, filters, accessible visualizations, export flows, offline-safe behavior, product analytics, and partner usability testing.

Core operating cadence

Cadence	Participants	Purpose
Weekly delivery review	All workstream leads	Evidence progress, blockers, data changes, and next seven-day commitments.
Biweekly partner clinic	Project lead + relevant city/operator leads	Resolve evidence requests and validate product decisions.
Monthly gate review	Project lead, partner sponsors, all accountable leads	Approve continuation, scope changes, risk treatment, and public outputs.

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Team structure · Climate, equity, partnerships, and assurance

Cross-functional review keeps the model useful without overstating the evidence.

Name	Pilot role	Relevant skills applied / accountable outputs
Andrei Dela Cruz	Climate and Grid Research Lead	DOE source review, grid/emission-factor interpretation, charging evidence framework, climate-method boundaries, and utility/depot refinement questions.
Russel Mendez	Equity and Community Engagement Lead	Inclusive workshop design, rider/community evidence, equity safeguards, contested-assumption log, accessibility needs, and validation of density-proxy limitations.
Tj Moreno	Operator and Pilot Partnerships Lead	Cooperative engagement, fleet and finance readiness rubric, operator overrides, partnership follow-up, implementation dependencies, and workshop logistics.
JM Palaganas	QA, Documentation and Demo Lead	Test plan, issue tracking, browser and data QA, methodology/readme maintenance, submission artifacts, training materials, and live-demo readiness.

Decision rights

Decision	Recommends	Approves	Required consultation
Data/source change	Data + domain lead	Project Lead	Affected workstreams
Score definition/weight	Analytics + equity leads	Monthly gate review	LGU, operator, rider/community participants
Public route shortlist	Product + policy leads	Partner sponsor + Project Lead	All evidence leads
Pilot scale decision	Project Lead	Partner sponsors	All workstreams and pilot users

Role assignments can be adjusted during mobilization, but every workstream, decision gate, and published artifact must retain one accountable owner.

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Risk management, handover, and scale path

The pilot treats evidence gaps as managed work—not as numbers to fill silently.

Risk	Level	Treatment	Owner
Historic or incomplete route data	High	Refresh priority routes with LGU/operator evidence; retain date and geometry confidence on every record.	Lead Data Engineer
Equity proxy misread as settlement truth	High	Use community validation; prohibit claims beyond density exposure; add missing-state language where needed.	Equity Lead
Grid score mistaken for local capacity	High	Pair every shortlist with depot/feeder questions; keep regional score constant and visibly coarse.	Climate/Grid Lead
Operator participation or finance evidence delayed	Medium	Keep neutral placeholder; schedule early clinics; document nonresponse; avoid forced precision.	Partnerships Lead
Model weights become political black box	Medium	Publish default and alternate scenarios; log who proposed changes and why.	Analytics Lead
API failure during demonstration	Low	Cache selected AI narratives and retain deterministic offline explanations for every ranked route.	Product Lead

Handover package

OPERATE Runbook, environment setup, data refresh, overrides, tests, and issue log	GOVERN Source ledger, weight decisions, public methodology, risk register, and named owners	SCALE Validated gaps, partnership map, 12-month roadmap, resource ask, and replication checklist
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End-of-pilot recommendation

Scale only if the refreshed route baseline, community review, depot feasibility, and operator evidence materially improve confidence in the shortlist—and if named partner teams can maintain the data and governance after the hackathon team steps back.

Prepared 14 August 2026 · Route2Zero Team